Teleoperating robots with virtual reality

A new virtual reality (VR) system from Massachusetts Institute of Technology (MIT) could make it easier for factory workers to telecommute. Developed by researchers from MIT's Computer Science and Artificial Intelligence Laboratory (CSAIL), the system lets users teleoperate a robot using an Oculus Rift headset. It embeds the user in a VR control room with multiple sensor displays, making it feel like they're inside the robot's head. By using hand controllers, users can match their movements to the robot's movements to complete various tasks.

Consisting of a headset and hand controllers, CSAIL's new VR system enables users to teleoperate a robot using an Oculus Rift headset (Photo: Jason Dorfman/MIT CSAIL)

"A system like this could eventually help humans supervise robots from a distance," says CSAIL postdoc Jeffrey Lipton, who was the lead author on a related paper about the system. "By teleoperating robots from home, blue-collar workers would be able to telecommute and benefit from the IT revolution just as white-collars workers do now."

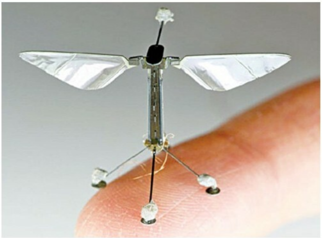
The researchers even imagine that such a system could help employ increasing number of jobless video-gamers by 'gamifying' manufacturing positions.

The team used the Baxter humanoid robot from Rethink Robotics, but said that the system can also work on other robot platforms and is also compatible with the HTC Vive headset.

Autonomous flying microrobots

Inspired by the biology of a bee, researchers at Harvard University's Wyss Institute are developing RoboBees—man-made systems that could perform myriad roles in agriculture or disaster relief. A RoboBee measures about half the size of a paper clip, weighs less than one-tenth of a gram, and flies using artificial muscles composed of materials that contract when a voltage is applied.

RoboBee is motivated by the idea to develop autonomous micro-aerial vehicles capable of self-contained, self-directed flight and of achieving coordinated behaviour in large groups. To that end, it consists of three main components: the body, brain, and colony. Body development consists of constructing robotic insects able to fly on their own with the help of a compact and seamlessly integrated power source. Brain development is concerned with smart sensors and control electronics that mimic the eyes and antennae



RoboBee (Source: Wyss Institute at Harvard University)